Machine Learning × Advanced Python · PGE 2

Predicting the final grade of secondary-school students

An early-warning tool for school counsellors: who needs extra support, and when?

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The data and one design choice

649

students, 30 features, no missing values

15

students with G3 = 0: probably dropout or skipped exam, kept on purpose

519 / 130

train and test, split once, seed 42

G1 and G2 are excluded. The tool must work early, before any grade exists. The price: a lower accuracy.

Preparation lives inside the pipeline

Numeric columns

Median imputation, then standardisation

Categorical columns

Most-frequent imputation, then one-hot encoding

Model

Dummy, linear, Ridge, Lasso, k-NN, Huber

Everything is re-fitted on the training part of each fold, so nothing from a validation fold leaks into the scaling.

5-fold cross-validation, repeated 2 times, same folds for every model.

Six models, same cross-validation

Model	Best hyperparameters	CV RMSE
Dummy (average grade)	none	3.19
Linear regression	none	2.76
Huber (chosen)	alpha 10, epsilon 1.35	2.71
Ridge	alpha 100	2.71
Lasso	alpha 0.05	2.73
k-NN	k 20, distance weights	2.81

Huber, Ridge and Lasso differ by under 0.02, far below the fold spread (about 0.3): a tie. Huber wins on its robust loss: the 15 zero grades cause a third of the squared error.

One test, on 130 held-out students

2.85

test RMSE (CV: 2.72)

2.03

test MAE (CV: 1.94)

3.19

Dummy baseline, CV RMSE

- The gap to cross-validation is smaller than the fold spread: no overfitting.
- Three large overpredictions, probably students with G3 = 0.
- Predictions stay between 7 and 15: extreme cases are not found.

What the model says, and where it stops

Main features

Past failures: −0.75 per standard deviation

Wants higher education: +0.55

Extra school support: −0.57

Associations, not causes

Two limits

Dropouts (G3 = 0) are not detected

649 students, two schools, error of about 2 points

Risks for real people

Unfair labels, proxies for social background, sensitive data

Supports the counsellor, never replaces

A package anyone can install and run

data.py DatasetLoader

features.py ColumnTransformer

models.py Trainer

cli.py train, evaluate, predict

14 tests, ruff, pinned versions

```
pytest
14 passed
student-grades predict --input examples/new_case.json
Predicted final grade (G3): 12.4 / 20
```